Computational Astrophysics Homework2

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1 Problem 1: Integrator Suite

1.1 Implementation

A common interface, orbit(acc, x_0, v_0, method=" "), was implemented to solve second-order equations of motion,

$$\ddot{\mathbf{x}} = \mathbf{a}(\mathbf{x}).$$

using four numerical methods: Euler, RK2, RK4, and leapfrog. RK2 and RK4 are applied to the corresponding first-order system in position and velocity, while leapfrog uses the kick-drift-kick (KDK) scheme. The interface returns position and velocity at the requested output times.

The implementation was tested with pytest. The tests covered convergence order for a harmonic oscillator, energy behavior under leapfrog integration, time reversibility for an eccentric Kepler orbit, and the number of acceleration evaluations for leapfrog.

All 12 tests passed, confirming the expected numerical behavior of the four integration methods.

2 Problem 2: Kepler Convergence Study

The circular Kepler orbit was simulated with $GM = 1$, $\mathbf{r}_0 = (1, 0)$, $\mathbf{v}_0 = (0, 1)$, and period $P = 2\pi$. Four methods, Euler, RK2, RK4, and leapfrog, were tested using $N = 100$, 200, 400, 800, 1600, and 3200 steps per orbit. The final position error was measured after one and ten orbits. The fitted slopes and pairwise Richardson estimates were calculated from these six step sizes.

2.1 Results

Table 1 compares the fitted slopes with the theoretical convergence orders. For one orbit, all slopes are within 0.2 of their expected values.

Table 1: Fitted convergence slopes.

Method	Expected order	One orbit	Ten orbits
Euler	1	0.8622	0.1499
RK2	2	2.0128	2.1251
RK4	4	4.0811	4.4101
Leapfrog	2	1.9996	1.9996

Figure 1 shows the error for one and ten orbits. The Richardson estimates in Table 2 provide a pairwise check of the convergence order.

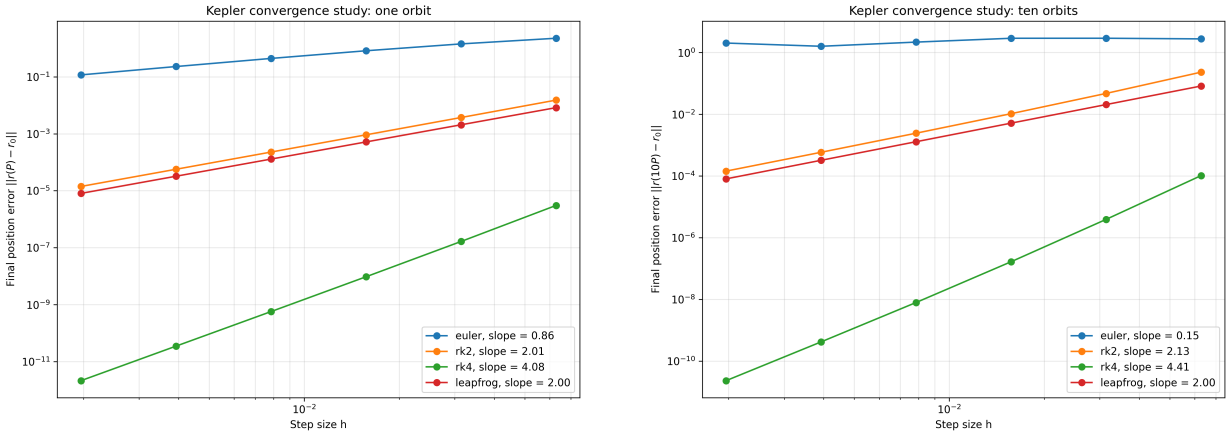


Figure 1: Final position error after one orbit (left) and ten orbits (right) versus step size on logarithmic axes. The fitted slopes are shown for each method.

Table 2: Pairwise Richardson estimates for one and ten orbits.

Orbits	Method	$100 \to 200$	$200 \to 400$	$400 \to 800$	$800 \to 1600$	$1600 \to 3200$
One	Euler	0.6590	0.7998	0.8941	0.9465	0.9733
One	RK2	2.0295	2.0188	2.0104	2.0055	2.0028
One	RK4	4.2038	4.1140	4.0606	4.0299	4.0246
One	Leapfrog	1.9984	1.9996	1.9999	2.0000	2.0000
Ten	Euler	-0.0688	0.0022	0.4128	0.4492	-0.3470
Ten	RK2	2.2901	2.1810	2.1014	2.0538	2.0277
Ten	RK4	4.7158	4.5586	4.3893	4.2407	4.1749
Ten	Leapfrog	1.9980	1.9996	1.9999	2.0000	2.0000

2.2 Discussion

For one orbit, the Richardson estimates approach the expected orders as the step size decreases. Euler approaches first-order convergence, while RK2, RK4, and leapfrog approach their expected second-, fourth-, and second-order behavior.

Over ten orbits, RK2 and leapfrog still show close to second-order convergence. However, Euler's error remains large and does not decrease steadily as the step size decreases. At coarse step sizes, the numerical orbit has deviated so far from the true circular orbit that the expected convergence pattern has not yet appeared. This is pre-asymptotic behavior.

The fitted slope of RK4 over ten orbits is 4.4101, above its expected order of four. This does not mean that RK4 has become a higher-order method. The fitted slope is affected by the limited range of step sizes used in the fit.

Additional RK4 runs with $N = 6400$ and 12800 steps per orbit were used to examine the behavior at smaller time steps. Over ten orbits, the error decreases from 2.32×10^{-11} at $N = 3200$ to 6.03×10^{-13} at $N = 6400$, but increases to 2.74×10^{-12} at $N = 12800$. The corresponding Richardson estimates change from 5.2687 to -2.1843. Over one orbit, the errors at $N = 6400$ and 12800 are 2.05×10^{-13} and 1.14×10^{-13}, respectively.

These results indicate that RK4 has reached the roundoff floor, where floating-point errors limit the accuracy and smaller time steps no longer guarantee smaller errors. RK4 is therefore the saturated method in these additional tests. The main fits use only the six required step sizes ($N = 100$ to 3200); the two additional resolutions are used only to examine roundoff effects.

3 Problem 3: Energy and Angular-Momentum Drift

3.1 Setup

Eccentric Kepler orbit with eccentricity $e = 0.5$ and semi-major axis $a = 1$. The initial conditions at apocenter were

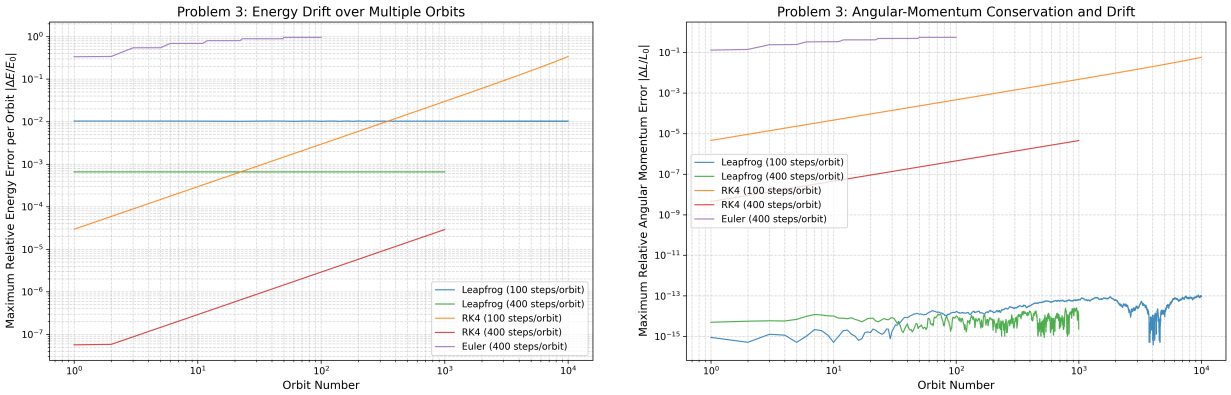
$$\mathbf{r}_0 = (1.5, 0), \qquad \mathbf{v}_0 = \left(0, \frac{1}{\sqrt{3}}\right).$$

The orbital period is $P = 2\pi$, with initial energy $E_0 = -1/2$ and angular momentum $L_0 = \sqrt{3}/2$.

Leapfrog and RK4 were run with 100 steps per orbit for 10,000 orbits and with 400 steps per orbit for 1,000 orbits. Forward Euler was run with 400 steps per orbit for 100 orbits. Energy and angular momentum were evaluated at every integration step. For each orbit, I recorded the maximum relative energy error, the end-of-orbit relative energy error, and the maximum relative angular-momentum error.

3.2 Energy and Angular-Momentum Errors

Figure 2 compares the maximum relative energy and angular-momentum errors within each orbit for the simulated integrator runs.



(a) Maximum relative energy error

(b) Maximum relative angular-momentum error

Figure 2: Maximum relative energy and angular-momentum errors within each orbit for the integrator runs.

Table 3 summarizes the energy and angular-momentum errors for the leapfrog and RK4 simulations.

Table 3: Energy and angular-momentum errors for leapfrog and RK4. Leapfrog energy errors are run-wide maxima; RK4 energy errors are measured at the last orbit.

Method	Steps/orbit	Energy error	Angular-momentum error
Leapfrog	100	1.0322×10^{-2}	1.0987×10^{-13}
Leapfrog	400	6.5716×10^{-4}	2.6024×10^{-14}
RK4	100	2.9825×10^{-2}	5.7186×10^{-2}
RK4	400	2.8988×10^{-5}	4.4850×10^{-6}

For leapfrog with 100 steps per orbit, the maximum relative energy error remained bounded at 1.0322×10^{-2} over 10,000 orbits, below the acceptance threshold of 2.0×10^{-2}. The maximum errors over orbits 1–500 and 9501–10000 were nearly identical, with a relative difference of only 0.000002%. This indicates that the energy error did not exhibit noticeable secular growth over the simulated interval.

With 400 steps per orbit, the maximum relative energy error over 1,000 orbits decreased to 6.5716×10^{-4}. The ratio of the two maximum errors was 15.71, close to the factor of 16 expected when the step size is reduced by a factor of four for a second-order method.

The RK4 results exhibit a different long-term behavior. The fitted log-log slopes of the end-of-orbit energy error over orbits 10–1000 were 1.0041 and 1.0000 for 100 and 400 steps per orbit, respectively. Both are consistent with approximately linear growth in the energy error with the number of orbits. At orbit 1000, the end-of-orbit energy error decreased from 2.9825×10^{-2} to 2.8988×10^{-5} when the number of steps per orbit increased from 100 to 400, a reduction factor of 1028.86. This is close to the factor of $4^5 = 1024$ expected from the observed fifth-power dependence of the accumulated error on the step size in this test.

The angular-momentum results also distinguish the two methods. For leapfrog, the maximum relative angular-momentum errors were 1.0987×10^{-13} over 10,000 orbits with 100 steps per orbit and 2.6024×10^{-14} over 1,000 orbits with 400 steps per orbit. Both are below 10^{-12}. For RK4, the corresponding maximum errors were 5.7186×10^{-2} and 4.4850×10^{-6}, respectively. These results show that leapfrog preserved angular momentum to near numerical precision in these runs, whereas the RK4 errors were larger, particularly at the coarser resolution.

3.3 Euler Trajectory

Figure 3 shows the trajectory obtained using the Forward Euler method with 400 steps per orbit over 100 orbits.

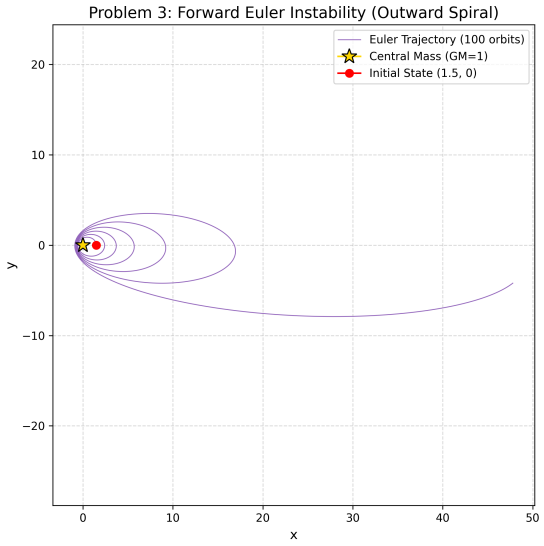


Figure 3: Forward Euler trajectory for the eccentric Kepler orbit.

The maximum relative energy error within the first orbit was 0.3359, exceeding the required threshold of 0.1. The trajectory spirals outward rather than remaining close to the original bound orbit. The maximum relative angular-momentum error over the full 100-orbit run was 5.5134×10^{-1}, further demonstrating the substantial numerical error accumulated by Forward Euler in this simulation.

3.4 Discussion

The results illustrate the distinction between the formal order of a numerical integrator and its long-term behavior for Hamiltonian systems. Although RK4 is a fourth-order method, it is not generally symplectic and does not guarantee bounded long-term energy error. In these simulations, its end-of-orbit energy error grew approximately linearly with the number of orbits, as indicated by the fitted log-log slopes close to one.

Leapfrog, by contrast, is a symplectic second-order method. Its energy error remained bounded over the simulated interval, and increasing the resolution reduced the maximum energy error by a factor consistent with second-order convergence. It also conserved angular momentum to near numerical precision in both runs. These results illustrate why a symplectic method can be advantageous for long-term orbital integrations even when its formal order is lower than that of RK4.

Finally, the Forward Euler trajectory spiraled outward, with a first-orbit maximum relative energy error of 0.3359. This behavior is consistent with its poor performance for long-term orbital integration in this test. Overall, the simulations demonstrate that both resolution and the geometric properties of an integration method affect the accuracy and stability of long-term orbital calculations.

4 Problem 4: Sun–Jupiter–Trojan Co-orbital Motion

4.1 Setup

I chose Option A, using the Problem 1 leapfrog integrator with direct-summation gravitational accelerations. In units with $G = 1$, total mass 1, and Sun–Jupiter separation 1, $\mu = m_J/(M_\odot + m_J) = 9.537 \times 10^{-4}$ and $P_J = 2\pi$. The Sun and Jupiter start at $\mathbf{r}_\odot = (-\mu, 0)$ and $\mathbf{r}_J = (1-\mu, 0)$, with velocities $(0, -\mu)$ and $(0, 1-\mu)$, respectively. The massless test particles were initialized at L4, $\mathbf{r} = (1/2 - \mu, \sqrt{3}/2)$, $\mathbf{v} = (-\sqrt{3}/2, 1/2 - \mu)$, and on the unit circle at 40° and 25°, with co-rotating velocities. All three runs covered 200 Jupiter orbits with $h = P_J/400$.

Positions were transformed to the frame rotating with Jupiter using its actual barycentric angle $\phi_J = \text{atan2}(y_J, x_J)$, rather than Ωt. I measured the relative longitude, barycentric radius, and closest approach in Hill radii, where $r_H = (\mu/3)^{1/3} \approx 0.0683$. The Jacobi energy was calculated from inertial-frame quantities:

$$E_J = \frac{1}{2}(v_x^2 + v_y^2) - \frac{1-\mu}{r_1} - \frac{\mu}{r_2} - (xv_y - yv_x),$$

where r_1 and r_2 are the distances to the Sun and Jupiter.

4.2 Trajectories and Conservation

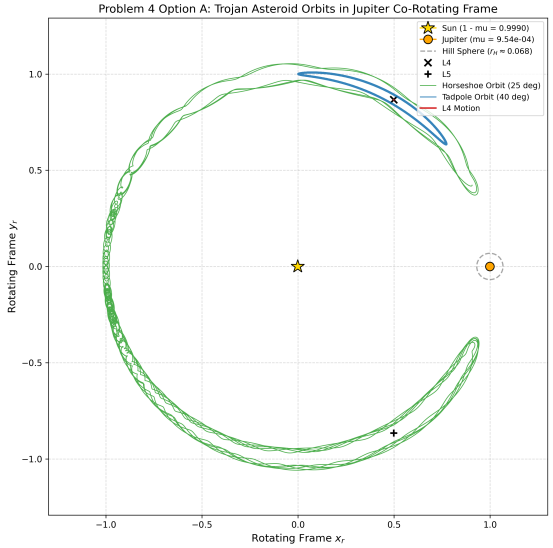


Figure 4: Test-particle trajectories in Jupiter's rotating frame.

The L4 particle stayed within 1.233037×10^{-4} of L4, satisfying the 10^{-3} tolerance. The 40° particle exhibited tadpole motion: its relative longitude ranged from 39.22° to 90.10°, its barycentric radius from 0.9738 to 1.0261, and its closest approach was $9.82 r_H$. The 25° particle followed a horseshoe orbit, with longitude ranging from 21.86° to 338.38°, crossing 180° and reaching the L5 side. Its closest approach was $5.49 r_H$, outside Jupiter's Hill sphere.

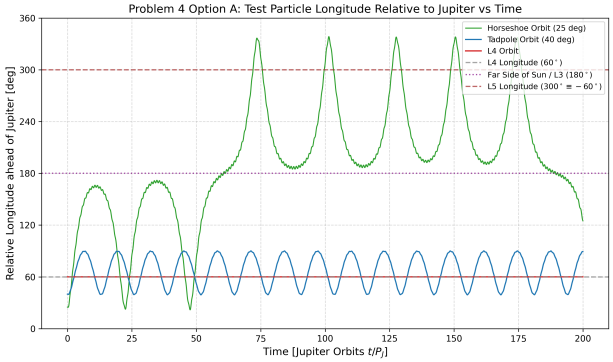


Figure 5: Relative longitude versus time in Jupiter orbital periods.

The longitude plot confirms these distinct behaviors: the L4 particle stayed near 60°, the 40° particle librated around L4, and the 25° particle traversed a wide angular range consistent with horseshoe motion. The 40° particle satisfied the required longitude range of 30°–100° and radial range of 0.96–1.04. The 25° particle passed the required closest-approach threshold of $4 r_H$ and never entered Jupiter's Hill sphere.

The maximum relative Jacobi-energy errors were 8.344256×10^{-8} (L4), 4.571451×10^{-7} (40°), and 1.226848×10^{-6} (25°), all below 10^{-5}. The Sun–Jupiter pair's maximum relative energy error was 1.521454×10^{-8}, below 10^{-6}.

4.3 Jupiter Phase Lag

The measured phase lags were:

Time step	Phase lag
$P_J/400$	$-5.9212°$
$P_J/1000$	$-0.9475°$

Since $h_1/h_2 = 1000/400 = 2.5$, second-order scaling predicts a phase-lag ratio of $2.5^2 = 6.25$. The measured ratio of the phase-lag magnitudes was 6.2495, differing from the prediction by only 0.008%, well within the required 20% tolerance.

4.4 Conclusion

The three simulations showed distinct co-orbital behaviors: the particle at L4 remained near the equilibrium point, the 40° particle exhibited tadpole motion, and the 25° particle followed a horseshoe trajectory. All three runs satisfied the specified trajectory, closest-approach, and Jacobi-energy criteria. The Sun–Jupiter energy error and the phase-lag test also met the required tolerances.